COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	*Information security lab*
Instructor	*Mrs. Ambreen Gull*

Student Details

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Task: 1

Write a Python function to calculate GCD using Euclid’s Algorithm

```python
def gcd_euclid(a, b):
    while b != 0:
        a, b = b, a % b
    return a

# Example
print("GCD:", gcd_euclid(48, 18))
```

Output:

```
C:\Users\HP\IdeaProjects\PythonProject\Pyth
GCD: 6

Process finished with exit code 0
```

Task: 2

GCD with Step DisplayModify the GCD function to display each step of Euclid’s Algorithm.

```python
def gcd_steps(a, b):
    print("Steps of Euclid's Algorithm:")
    while b != 0:
        print(f"{a} = {b} * ({a//b}) + {a%b}")
        a, b = b, a % b
    print("GCD =", a)
    return a

# Example
gcd_steps(48, 18)
```

Output:

```
C:\Users\HP\IdeaProjects\PythonProject\PythonPro
Steps of Euclid's Algorithm:
48 = 18 * (2) + 12
18 = 12 * (1) + 6
12 = 6 * (2) + 0
GCD = 6

Process finished with exit code 0
```

<u>Task: 3</u>

Recursive Implementation (5 Marks)
Implement Euclid’s Algorithm using recursion instead of a loop.

```
def gcd_recursive(a, b):  2 usages
    if b == 0:
        return a
    return gcd_recursive(b, a % b)

# Example
print("GCD (Recursive):", gcd_recursive( a: 48,  b: 18))
```

Output:

```
C:\Users\HP\IdeaProjects\PythonProject\PythonProje
GCD (Recursive): 6

Process finished with exit code 0
```